Watchful waiting versus methylprednisolone or polidocanol injections as treatment of simple bone cysts

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Aims

Simple bone cysts (SBCs) are among the most prevalent benign cystic bone lesions in children. The use of watchful waiting as management is debated, as are the type and timing of injection therapy. This study aims to compare clinical outcomes of watchful waiting and delayed and immediate percutaneous procedures in children with SBCs, together with the effectiveness of methylprednisolone and polidocanol.

Methods

All eligible paediatric patients with SBCs and minimal one year of follow-up were included from our tertiary musculoskeletal oncology referral centre (January 2000 to December 2024). Outcomes of interest were cumulative incidence of (partial) healing (CI-Healing), cumulative incidence of failure (CI-Failure), and prognostic factors for healing and failure. For CI-Healing, competing risk analysis was performed with failure as the competing event. For CI-Healing and CI-Failure of methylprednisolone and polidocanol, landmark analysis was performed. Prognostic factors were estimated using a multivariate cause-specific hazard model.

Results

In total, 160 paediatric patients with SBCs were included (mean age 8.7 years (SD 3.6); 128 (80%) males). Management consisted of watchful waiting (n = 41) and delayed (n = 86) or immediate (n = 33) percutaneous procedure. The median follow-up was 3.4 years (IQR 2.2 to 5.4). CI-Healing was significantly higher in immediate percutaneous procedures than in watchful waiting and delayed percutaneous procedures. Additionally, polidocanol performed significantly better than methylprednisolone regarding CI-Healing and CI-Failure. Lower limb localization was a significant negative prognostic factor for healing (cause-specific hazard ratio 0.38; 95% CI 0.18 to 0.82).

Conclusion

For treatment of SBCs, watchful waiting showed high chances of necessitating subsequent treatment, leading to lower CI-Healing and prolonged exposure to risk of failure. If treatment is required, polidocanol is a superior injectable compared with methylprednisolone, in terms of both healing and failure prevention.

Take home message

- For the treatment of simple bone cysts, polidocanol is a superior substance to methylprednisolone in terms of cumulative incidence of (partial) healing and failure.
- Watchful waiting often necessitates treatment at a later stage, but remains a

possible management strategy in patients with low fracture risk.

Introduction

Simple bone cysts (SBCs), also known as solitary, juvenile, or unicameral bone cysts, are benign cystic tumour-like lesions. SBCs typically are unicameral but can be septated. They are filled with clear, colourless to straw-coloured fluid, which post-traumatically can be serosanguineous or haemorrhagic.[1]

About 85% of SBCs are diagnosed in children aged three to 14 years, with a mean age of nine years at diagnosis,[2] and a 2 to 3:1 male predominance.[3] The true incidence of SBCs remains unclear, as they are often asymptomatic and found coincidentally. Symptoms can include pain, abnormal gait, functional limitation, and pathological fracture, which 63% to 87% have as first presentation.[4] Approximately 10% to 15% of SBCs show spontaneous healing after fracture.[5] However, fractured cysts can lead to growth arrests, deformity, and potentially osteonecrosis.[1]

Treatment aims are reducing potential symptoms and fracture risk, and promoting cyst healing. In clinical practice, primary management of SBCs is often watchful waiting, which entails periodical observation of the SBC with radiographs. This management strategy is used in various localizations, partly due to previous studies showing the safety and possibility of spontaneous healing, especially after (undisplaced) fracture.[5-7] Depending on fracture risk, activity restrictions may be imposed. In the case of impending fracture, e.g. due to large cyst volume or a highly active child, treatment may be necessary. Percutaneous procedures, e.g. injection with corticosteroids, bone marrow, bone substitutes, or sclerosing agents, are considered minimally invasive options.[7-9] Curettage, possibly with chemical adjuvants and/or filling, was historically the most common treatment, but is now mainly reserved for more aggressive and persistent cases, sometimes with additional internal fixation.[10] Internal fixation is often used in the case of unstable fractures, or pre-emptively to prevent an impending fracture from breaking. It has also been suggested as viable standalone treatment.[7]

Currently, clinical practice is highly variable without clear guidelines for surgical interventions. Multiple factors influence treatment choice, including cyst volume, symptom presence, localization, fracture (risk), local protocols, and physician preference.[7,11] However, there is limited evidence regarding the optimal management strategy. Previous studies are often small, retrospective, and focused on surgical treatment, overlooking patients who receive percutaneous procedures or watchful waiting.[9,12]

This study aims to compare cumulative incidence of failure and (partial) healing in children with SBCs managed through watchful waiting and delayed or immediate percutaneous procedures, and to evaluate differences between methylprednisolone (Depo-Medrol) and polidocanol (Aethoxysklerol) treatments. Through additional analysis into prognostic factors, we strive to enhance clinicians' ability to make informed decisions regarding the management of SBCs.

Methods

In this observational cohort study, children diagnosed with SBC at the Leiden University Medical Centre, Leiden, Netherlands, between January 2000 and December 2024 were included if they met the following criteria: age at diagnosis ≤ 16 years; SBC at any anatomical localization; and minimum follow-up of one year. Patients were excluded in the case of secondary cystic lesions and other known bone disorders (e.g. osteogenesis imperfecta, morbus Ollier, rickets), due to associated increased fracture risk. Diagnosis was established based on biopsy and/or typical radiological imaging.

Included patient data were collected from medical records and stored in a pseudoanonymized Castor EDC database.[13] Demographic, diagnostic, and treatment characteristics, including type of injectable solution, were extracted. In our centre, the standard injectable substance changed in 2011, so methylprednisolone was used from 2000 to 2015 (n = 1, 5.6% after 2011) and polidocanol from 2003 to 2024 (n = 4, 5.4% before 2011). If failure had occurred before first injection, that case was excluded from the injectable substance analyses. Index procedures were categorized: watchful waiting, immediate percutaneous procedures, delayed percutaneous procedures (i.e. percutaneous procedure after an initial six months minimum of watchful waiting, switched for any reason), open procedures, and internal fixation alone.

Imaging was analyzed at diagnosis, at time of treatment(s), and at different follow-up moments, focusing on cyst volume and degree of healing based on the modified Neer classification (Supplement 1).[14,15] Neer scores were assigned by a trained researcher and healing was defined as ≥ 50% healing with thick cortices.

Cyst size and volume were approximated on radiographs, using an established measuring protocol used for previous European Paediatric Orthopaedic Society studies.[7] Thereafter, cyst volume (V) was estimated using the formula for ellipsoid volume: $V = \frac{4}{3} \times \pi \times r_{AP} \times r_{CC} \times r_{ML}$, with the radii ($r$) determined by halving the maximal anteroposterior (AP) × craniocaudal (CC) × mediolateral (ML) distance in millimetres.

Activity restriction time was defined as the number of days a child could not play or do sports without restrictions.

Outcomes of interest were cumulative incidence of (partial) healing (CI-Healing), cumulative incidence of failure (CI-Failure), and prognostic factors for (partial) healing and failure in index groups, as well as between injection types (polidocanol and methylprednisolone). The definition of (partial) healing was modified Neer score A or B during follow-up. Failure was defined as pathological fracture of the cyst during follow-up, open (re)procedure or internal fixation after the index procedure, or > three percutaneous procedures (not mutually exclusive).

According to our institutional review board (2024-013) and Dutch law, this observational cohort study was not subject to the Medical Research Involving Human Subjects Act (G19-064) and need for patient informed consent was waived.

Statistical analysis

Descriptive analyses were performed for the included cohort and stratified for index procedures. Continuous normally distributed data were reported as means and SDs, while data with skewed distributions were summarized through medians and IQRs. Categorical data were presented as frequencies and percentages. The chi-squared test was employed to assess

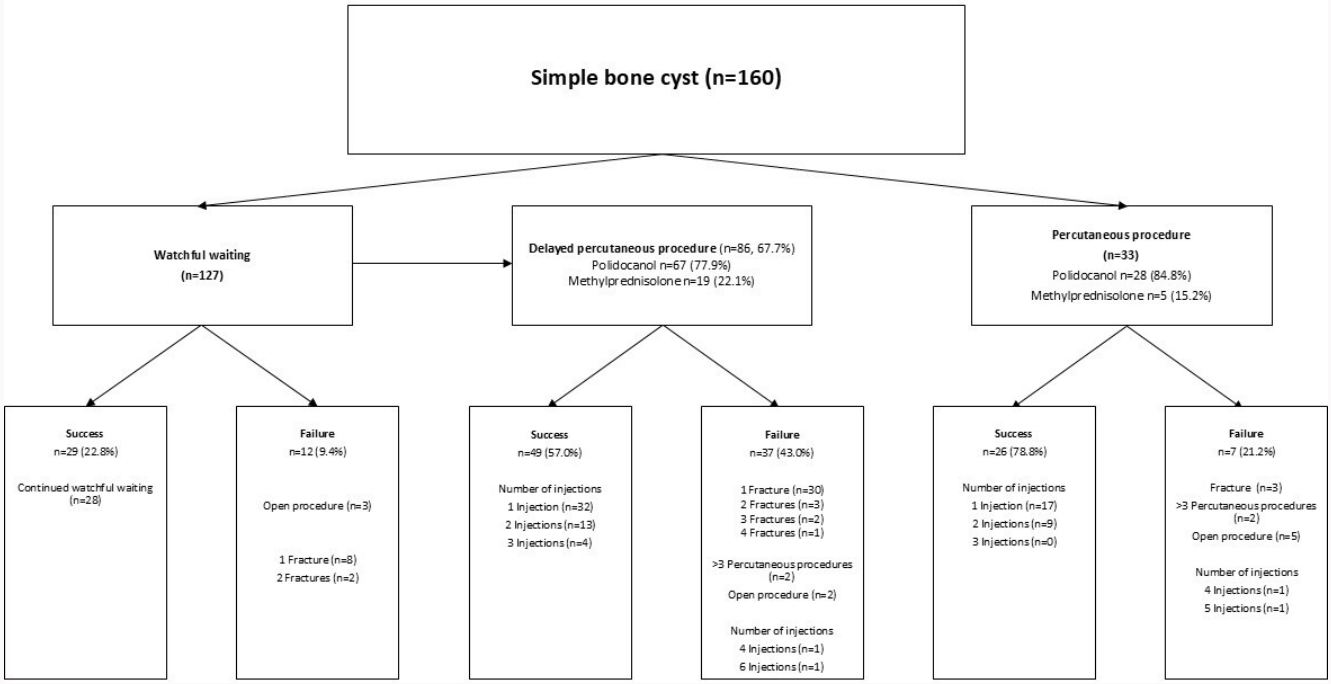


Fig. 1
Flowchart of index procedures and their successes and failures. Reasons for failure are not mutually exclusive.

differences between proportions. Fisher's exact test was used for rare events.

To assess the difference between groups for categorical and continuous variables, chi-squared and one-way analysis of variance tests were used. In the case of deviation from the normality assumption, non-parametric tests were employed.

Competing risks models were used to estimate CI-Healing and CI-Failure from diagnosis and from injection.[16] To assess the difference between the cumulative incidence for the index procedure and for injection type, Gray's test was used.[17] A landmark analysis with landmark point at injection time was performed to estimate the CI-Failure and CI-Healing for polidocanol or methylprednisolone.[18]

To estimate the effect of prognostic factors such as index procedure, injection type (for landmark analysis only), volume at diagnosis, fracture at diagnosis, cortical thickness < 1 mm, and upper or lower limb on the competing events of healing and failure, cause-specific hazard models were estimated and described as cause-specific hazard ratios (HR_{CS}) with 95% CIs.

A p-value < 0.05 was considered statistically significant. Competing risk analyses were performed in RStudio v. 2025.05.0 (Posit, USA) environment using the cmprsk package.[19] Remaining statistical analyses were performed using SPSS Statistics v. 29.0.2.0 (IBM, USA).

Results

In total, 173 patients with SBCs were identified and treated using the following strategies: delayed percutaneous procedure (n = 86); watchful waiting (n = 41); immediate percutaneous procedure (n = 33); internal fixation (n = 12); or open procedure (n = 1). A small subgroup of patients underwent internal fixation (n = 12) or open curettage (n = 1). These patients were excluded from the present comparative analyses, as these invasive strategies were specifically indicated for a distinct clinical subgroup with impending or actual pathological fractures, representing a different therapeutic pathway. To maintain a clinically homogeneous study population focused on conservative and percutaneous management strategies, these cases are described separately as real-world data in Supplement 2. Accordingly, 160 patients were included (Figure 1). Mean age at diagnosis was 8.7 years (SD 3.6) and 80% (n = 128) were male (Table I; missing data presented in Table II). The proximal humerus was the most common anatomical site (n = 97; 60.6%), followed by the proximal femur (n = 25; 15.6%; Figure 2). Fracture was the presenting symptom in 111/116 (95.7%) of upper limb and 20/44 (45.5%) of lower limb SBCs (p < 0.001, chi-squared test), and pain in 4/116 (3.4%) and 16/44 (36.4%; p < 0.001). The remainder, 1/116 (0.9%) and 8/44 (18.1%), were accidental findings (p < 0.001, Fisher's exact test).

In all patients, diagnosis was established using radiographs, with some patients receiving an additional CT scan (n = 15; 8.7%), MRI (n = 104; 60.1%), biopsy (n = 23; 13.3%), cytology (n = 2; 1.2%), and/or bone density scan (n = 1; 0.6%) in various combinations.

The median follow-up was 3.4 years (IQR 2.2 to 5.4). The IQR of activity restriction time was 0 to 6.3 months overall, 0 to 7.9 months for watchful waiting, 0 to 4.9 months for delayed percutaneous procedure, and 0.6 to 9.9 months for immediate percutaneous procedure.

The main complications after polidocanol injection were local skin reactions with swelling and pain (n = 1) or blisters (n = 1), generalized pruritus (n = 1), and perioperative tension drops (n = 1). No complications were seen after methylprednisolone injection.

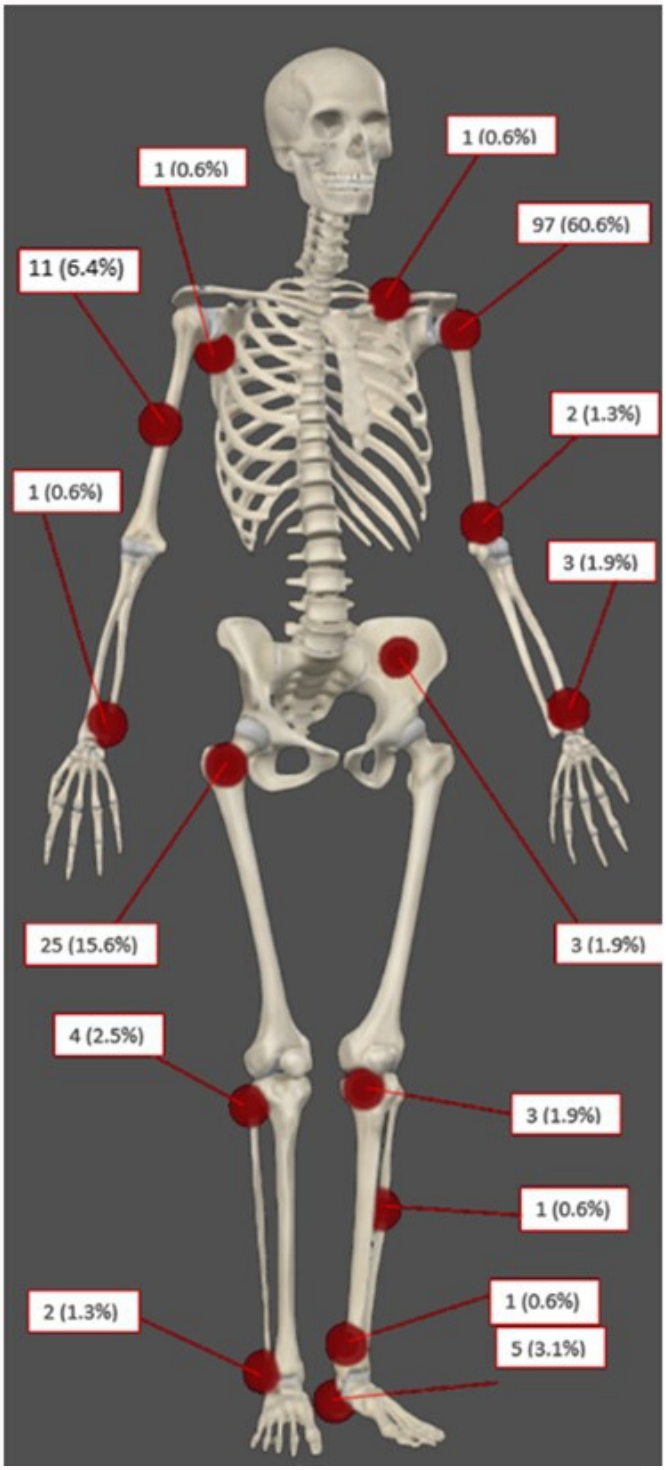


Fig. 2
Visualization of simple bone cyst localizations included in this study. Image courtesy of Complete Anatomy.

Failure

The overall failure rate was 56/160 (35.0%). Failure rates were 12/41 (29.3%) for watchful waiting, 37/86 (43.0%) for delayed percutaneous procedure, and 7/33 (21.2%) for immediate percutaneous procedure (Figure 1). Failure rates of injectable solutions were 7/18 (38.9%) for methylprednisolone and 11/63 (17.5%) for polidocanol.

CI-Failure per treatment strategy was 26% (95% CI 12 to 40) at two years and 30% (95% CI 14 to 46) at five years for watchful waiting, 32% (95% CI 22 to 42) at two years and 40% (95% CI 29 to 50) at five years for delayed percutaneous procedure, and 17% (95% CI 3 to 31) at two years and 24% (95% CI 8 to 41) at five years for immediate percutaneous procedure (p = 0.112; Figure 3).

Comparing between polidocanol (n = 74) and methylprednisolone (n = 18), CI-Failure was 12% (95% CI 4 to 20) at two years and 14% (95% CI 5 to 22) at five years, and 46% (95% CI 19 to 73) at two years and 46% (95%CI 19 to 73) at five years, respectively (p = 0.004; Figure 4).

Healing

CI-Healing for treatment strategies was 16% (95% CI 4 to 28) at two years and 40% (95% CI 19 to 62) at five years for watchful waiting, 24% (95% CI 15 to 33) at two years and 39% (95% CI 28 to 50) at five years for delayed percutaneous procedure, and 42% (95% CI 24 to 60) at two years and 76% (95% CI 52 to 99) at five years for immediate percutaneous procedure (p = 0.019; Figure 5).

For injection types, polidocanol had a CI-Healing of 51% (95% CI 38 to 63) at two years and 76% (95% CI 64 to 81) at five years, while methylprednisolone had 24% (95% CI 3 to 44) at two years and 39% (95% CI 3 to 75) at five years (p = 0.016; Figure 6).

Out of 100 patients who reached (partial) healing during follow-up, one patient (1%) had a subsequent fracture caused by high-energy trauma, 7.8 years after healing was achieved.

Prognostic factors

Results from a multivariate cause-specific proportional hazards regression model for prognostic factors of healing and failure are presented in Tables III and IV for index procedures and injection types, respectively. Significant prognostic factors for healing were lower limb cyst localization (HR_{CS} 0.38; 95% CI 0.18 to 0.82) and treatment with immediate percutaneous procedure (HR_{CS} 4.26; 95% CI 1.93 to 9.36). For failure, the use of polidocanol was a significant prognostic factor (HR_{CS} 0.35; 95% CI 0.13 to 0.91).

Discussion

This constitutes one of the largest studies to date on the treatment of SBCs in children, showing that watchful waiting is a viable option, although it often requires treatment at a later stage. Additionally, in the treatment of SBCs, polidocanol is a superior substance to methylprednisolone in terms of CI-Healing and CI-Failure.

There was a significant difference in CI-Healing between watchful waiting and both percutaneous procedure groups, with superior results for the immediate percutaneous procedures group. Our five-year CI-Healing of 76% for immediate percutaneous procedures was comparable to previous research on healing of bone cysts after methylprednisolone injection by Capanna et al[8] and Kadhim et al,[20] who found 80% and 77.4% (partial) healing rates, respectively, and was higher than the 65% rate of Ruiz-Arellanos et al.[21]

For watchful waiting, the five-year CI-Healing of 40% in our cohort was low compared with the 95.7% healing rate after 3.1 years of Green et al.[5] However, since they did not perform a competing risks analysis, comparisons should be made carefully. Additionally, differences in healing definition

Table I. Patient characteristics at baseline per treatment strategy, and for the complete cohort.

Characteristic	Watchful waiting (n = 41)	Delayed percutaneous procedure (n = 86)	Immediate percutaneous procedure (n = 33)	Total group (n = 160)
Demographics				
Male, n (%)	35 (85.4)	73 (84.9)	20 (60.6)	128 (80.0)
Mean age at diagnosis, yrs (SD)	9.5 (3.8)	7.9 (3.2)	9.9 (4.1)	8.7 (3.6)
Median follow-up, yrs (IQR)	2.5 (1.9 to 3.8)	3.9 (2.5 to 5.9)	3.3 (2.2 to 5.1)	3.4 (2.2 to 5.4)
Cyst characteristics				
Mean height, mm (SD)	52.9 (29.6)	56.6 (22.6)	55.9 (27.6)	55.6 (25.5)
Mean width, mm (SD)	23.6 (8.3)	23.2 (6.8)	30.0 (11.0)	24.7 (8.6)
Mean depth, mm (SD)	21.4 (6.9)	23.1 (7.0)	25.9 (8.6)	23.3 (7.5)
Median estimated volume, cm^3 (IQR)	16.1 (6.9 to 32.7)	16.9 (9.2 to 31.1)	21.6 (10.2 to 49.3)	17.9 (8.6 to 31.7)
Median minimal cortical thickness, mm (IQR)	0.8 (0.5 to 1.2)	0.7 (0.5 to 1.0)	1.0 (0.7 to 1.4)	0.8 (0.6 to 1.0)
Median closest distance to physis, mm (IQR)	6.5 (3.4 to 33.1)	4.2 (1.6 to 15.4)	14.8 (6.2 to 31.5)	6.5 (2.2 to 23.4)
Localization, n (%)				
Upper limb	30 (73.2)	75 (87.2)	11 (33.3)	116 (72.5)
Lower limb	11 (26.8)	11 (12.8)	22 (66.7)	44 (27.5)
Presenting symptoms, n (%)*				
Accidental discovery	5 (12.2)	1 (1.2)	4 (12.1)	10 (6.3)
Fracture at diagnosis	36 (87.8)	80 (93.0)	14 (42.4)	130 (81.3)
Pain	1 (2.4)	5 (5.8)	14 (42.4)	20 (12.5)
Abnormal gait†	1 (2.4)	2 (2.3)	4 (12.1)	7 (4.4)

*Presenting symptoms are not mutually exclusive.
†Abnormal gait was accompanied by pain in all cases.

("some consolidation" vs ≥ 50% healing with thick cortices in our study) may have influenced the difference.

We found a significant difference in CI-Healing between methylprednisolone and polidocanol (39% vs 76% at five years, p = 0.016), indicating the latter's superiority. As polidocanol has not been described separately in previous studies on SBC treatment, we have no comparison. However, participants in the methylprednisolone arm in the study by Wright et al[9] healed faster than ours, with a healing rate of 42% at two years compared with our CI-Healing of 24% at two years. This difference may have been caused by their more aggressive treatment protocol of three consecutive injections within one year.

Since the aim of treatment is maintaining stability while minimizing symptoms, activity restrictions, and complications, cyst healing is not the most interesting clinical outcome. However, our 1% fracture rate after achieving (partial) healing suggests that these patients have a strongly reduced risk of fracture and may be discharged from follow-up after healing.

In our study, there was a significant difference in CI-Failure between polidocanol and methylprednisolone injections. Traub et al[22] had a failure rate of 27.3% in their watchful waiting group and 36.6% after methylprednisolone injection. However, our results cannot be compared with their results, since a competitive risk model was not used.

It should be noted that fracture was included in our definition of treatment failure. However, in localizations such as the humerus, it may be considered to eventually promote the healing process, provided it does not lead to lasting deformities or neurovascular damage. Additionally, we did not classify recurrence/progression as failure, as these only become clinically relevant when symptomatic.

For failure, only the injectable substance used was found to be a significant prognostic factor, with less failure associated with polidocanol use. Previous studies have found age under five years,[8] age under ten years,[7] and cyst volume > 84.3 cm^3 (which, using the formula for ellipsoid volume, would be > 44.0 cm^3) to be prognostic factors.[22] Additionally, thin cortices and lower limb lesions are frequently cited as increasing fracture risk.[23] We could not reproduce these findings.

For healing, lower limb localizations were a significant negative prognostic factor (HR_{CS} 0.38), possibly because these cysts, due to different mechanical forces, present earlier in their natural history, often through symptoms of pain or abnormal gait (36.4% vs 3.4%) rather than fracture (45.5% vs 95.7%) for lower and upper limb, respectively. This suggests

Table II. Missing data at different timepoints.

Missing data, n (%)	Total group (n = 160)
Height cyst AD	3 (1.9)
Width cyst AD	3 (1.9)
Depth cyst AD	3 (1.9)
Estimated volume AD	3 (1.9)
Estimated volume at 6 mths	64 (40.0)
Estimated volume at 1 yr	57 (35.6)
Estimated volume at 2 yrs	61 (38.1)
Estimated volume at time of last follow-up	5 (3.1)
Minimal cortical thickness AD	4 (2.5)
Closest distance to physis AD*	9 (5.6)
Activity restriction time	17 (10.6)

*From nine with missing distance to physis, five were calcaneal lesions.
AD, at time of diagnosis.

Table III. Multivariate cause-specific proportional hazards of prognostic factors for healing and failure.

Prognostic factor	Failure, HR_{cs} (95%CI)	Healing, HR_{cs} (95%CI)
Pathological fracture at diagnosis		
Yes	0.91 (0.30 to 2.70)	1.40 (0.55 to 3.52)
No	Ref	Ref
Anatomical site		
Lower limb	0.61 (0.27 to 1.40)	0.38 (0.18 to 0.82)*
Upper limb	Ref	Ref
Cortical thickness, mm		
< 10	1.31 (0.72 to 2.38)	1.64 (0.91 to 2.94)
≥ 10	Ref	Ref
Age, yrs		
≤ 10	N/A	0.81 (0.45 to 1.44)
> 10	N/A	Ref.
Volume (cm^3)	0.99 (0.98 to 1.01)	0.99 (0.98 to 1.00)
Treatment group		
Watchful waiting	Ref	Ref
Delayed percutaneous procedure	1.35 (0.67 to 2.69)	1.18 (0.61 to 2.31)
Immediate percutaneous procedure	0.81 (0.29 to 2.22)	4.26 (1.93 to 9.36)*

*Statistical significance.
HR_{cs}, cause-specific hazard ratio; N/A, not applicable; Ref, reference category.

Table IV. Multivariate cause-specific proportional hazards of prognostic factors for healing and failure of landmark analysis based on injection type.

Prognostic factor	Failure HR_{cs} (95%CI)	Healing HR_{cs} (95%CI)
Injected substance		
Polidocanol	0.35 (0.13 to 0.91)*	2.28 (0.90 to 5.77)
Methylprednisolone	Ref	Ref
Injection timing		
Immediate	0.98 (0.37 to 2.55)	1.00 (0.57 to 1.74)
Delayed	Ref	Ref

*Statistical significance.
HR_{cs}, cause-specific hazard ratio; Ref, reference category.

they are more often in their active phase, and might take longer to heal.

Immediate percutaneous procedures were associated with significantly better healing than watchful waiting. However, when corrected for covariates, this effect disappeared. This change was partly due to failures leading to the exclusion of delayed percutaneous treatment from the landmark analysis, and it may suggest that the observed association is influenced more by the substance used than by the injection timing.

SBCs are often suggested to have self-healing tendencies post-traumatically.[6] However, we found no significant difference in healing after fracture. This finding is supported by Cha et al,[24] who reported no difference in healing rate or time to healing.

When considering different treatment options, both cyst and child characteristics must be considered, alongside patient and parent preferences.

Immediate percutaneous procedures are a reliable option, as they are superior in CI-Healing and CI-Failure. Furthermore, it should be noted that in our study, delayed percutaneous procedure was not a treatment choice, but rather already a form of failure following watchful waiting. Therefore, the CI-Failure of watchful waiting may be considered higher and the CI-Healing lower than presented. Nonetheless, watchful waiting may still be applied in cases with low fracture risk, although the considerable burden of prolonged restrictions in sports and play must be considered. In our study, we did not find significant differences in median activity restriction time, although this is most likely due to confounding by indication. When discussed with parents and patients, the high risk of future intervention must be mentioned.

To the best of our knowledge, ours is the second study describing the use of polidocanol in SBCs, after a previous study by the EPOS study group.[7] Polidocanol is usually used as a sclerosing agent in varices or aneurysmal bone cysts. However, this approach is not generally mentioned for SBCs in the available literature.[3,4,10] Multiple possible mechanisms of action can be formulated, depending on the aetiology to which one adheres. These include the sclerosing of vessels around the cyst that lead to the high intramedullary pressure (venous drainage blockage theory),[25,26] or cyst wall disruption, with discrete inflammation, leading to bone remodelling (secondary to trauma, local growth disturbance, or genetic translocation).[2,27,28]

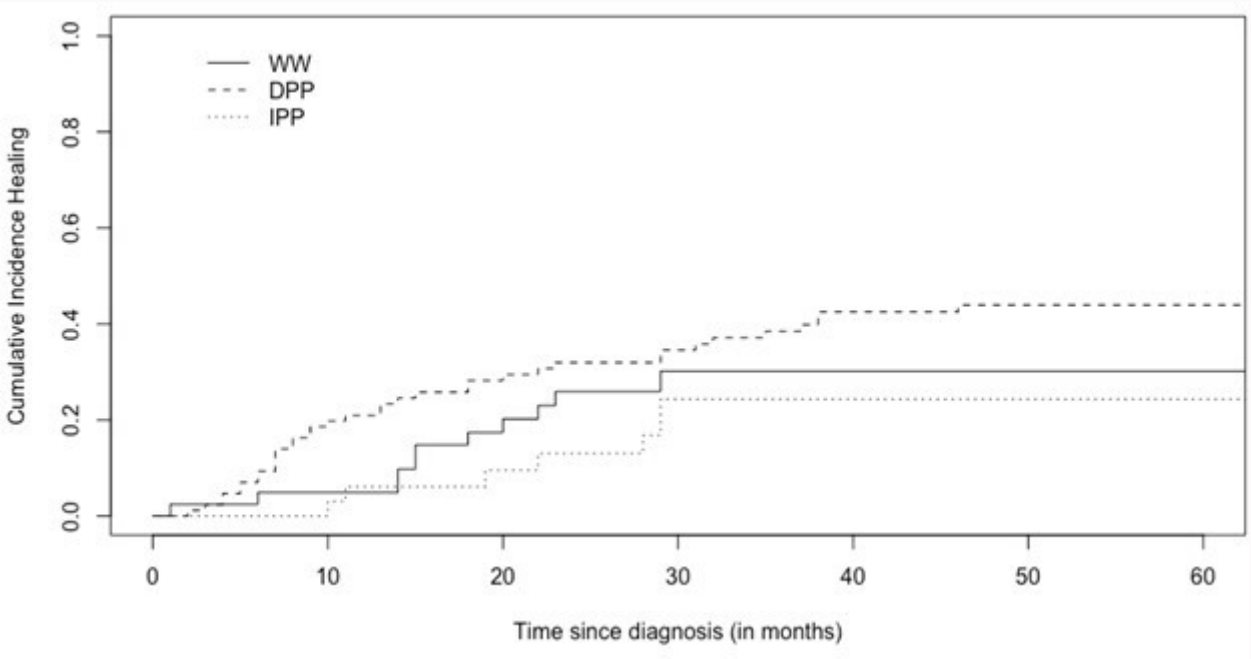


Fig. 3
Cumulative incidence of failure per index procedure. DPP, delayed percutaneous procedure; IPP, immediate percutaneous procedure; WW, watchful waiting.

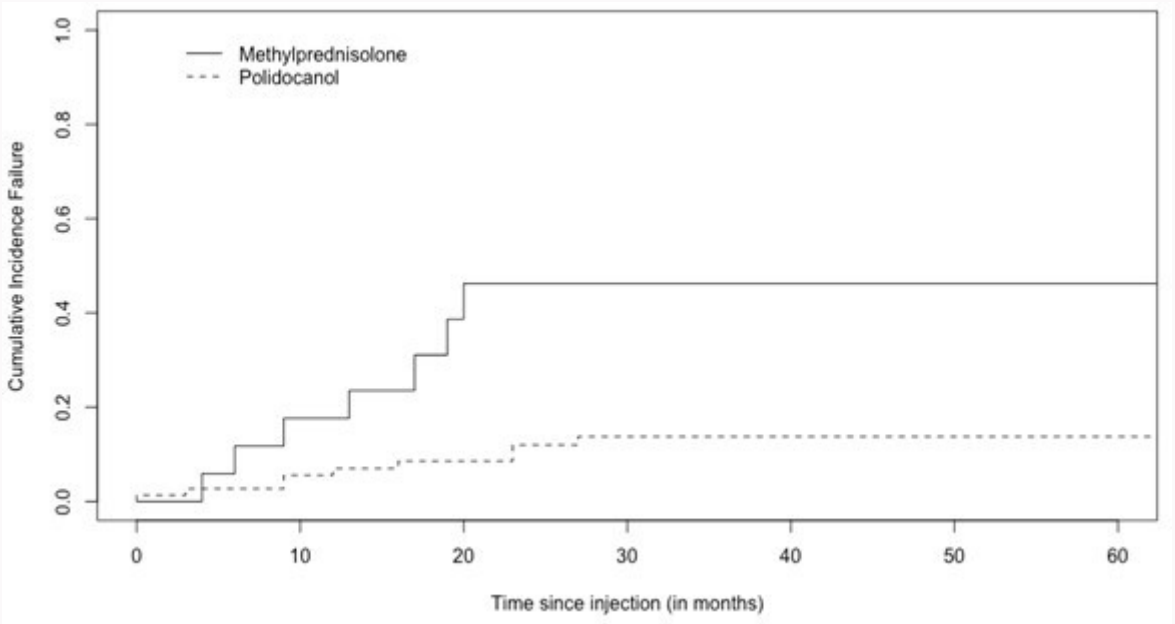


Fig. 4
Cumulative incidence of failure per injection type.

Since polidocanol showed better CI-Failure and CI-Healing, it does appear to be a promising alternative to methylprednisolone injections. The use of polidocanol may decrease the morbidity of multiple injections by producing faster healing rates, which has previously been stated as the biggest downside of methylprednisolone injections.[29] However, local skin reactions and rare complications, such as intraoperative tension drops and possible anaphylactic shocks, should be heeded.[30]

This is one of the first and largest studies to describe different treatment strategies in a unique cohort of patients with SBCs. Furthermore, including analyses of both healing and failure is advantageous, as these outcomes clearly interact and are not simply opposites. By using a multivariate cause-specific proportional hazards regression model, we were able to estimate true hazard ratios more precisely than previous studies, which only used univariate models. In addition to evaluating treatment options, we assesed both the question

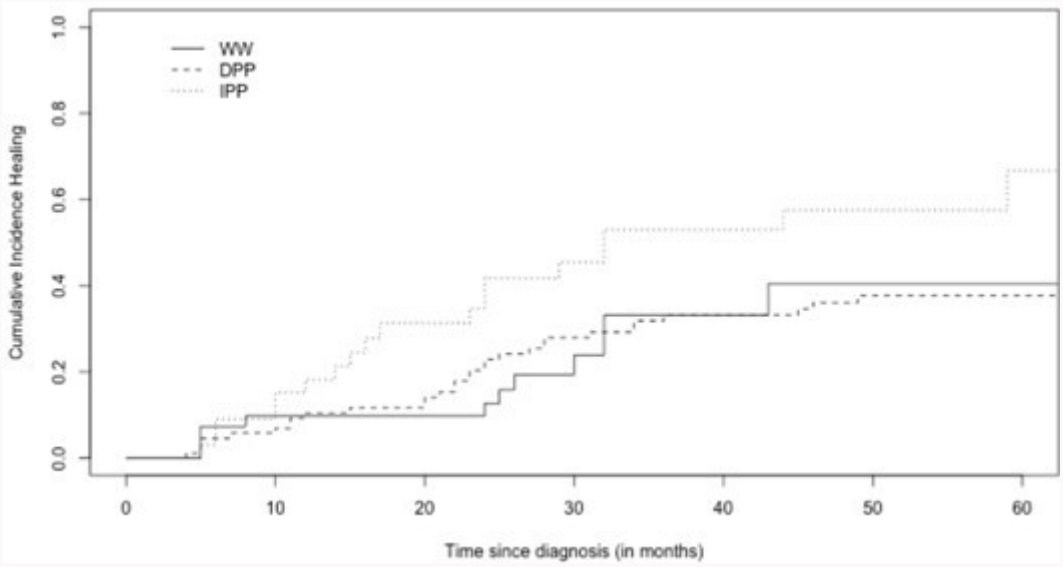


Fig. 5
Cumulative incidence of (partial) healing per index procedure. DPP, delayed percutaneous procedure; IPP, immediate percutaneous procedure; WW, watchful waiting.

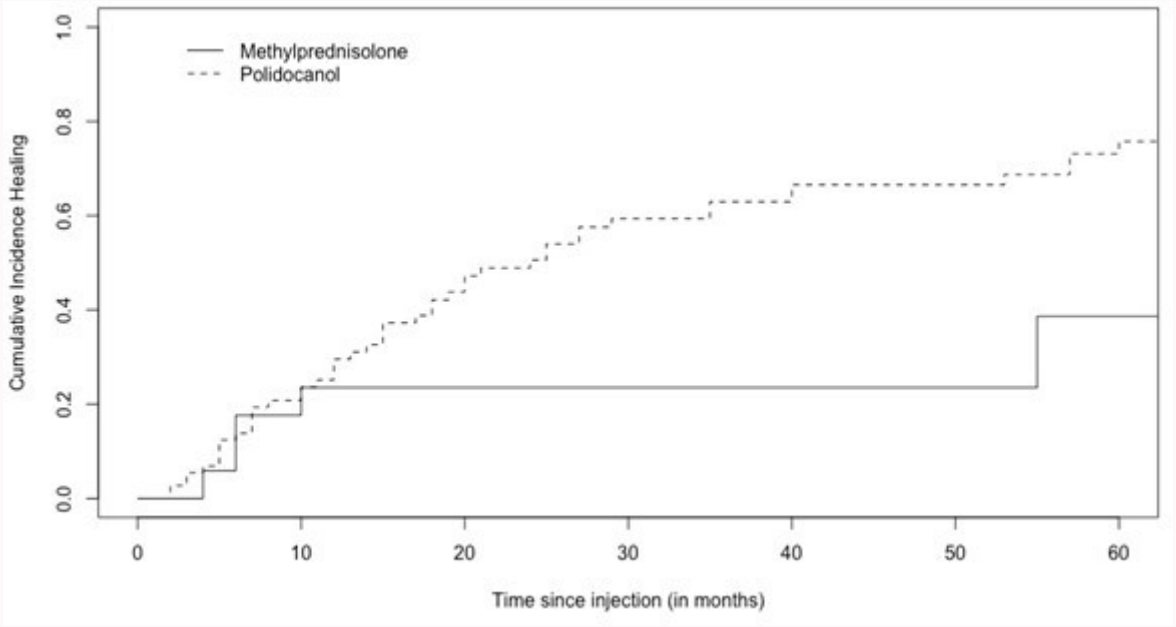


Fig. 6
Cumulative incidence of (partial) healing per injection type.

of superior treatment modality, and the question of treatment timing. Lastly, we present a large cohort of patients treated with polidocanol, suggesting that this injection substance may be a promising alternative to methylprednisolone, potentially reducing the need for more invasive curettage.

Indication bias is a major limitation of our observational study design, with the immediate percutaneous procedures group being older, with cysts generally further from the physis and more often in the lower limb. However, through our multivariate model, we were able to correct for most confounders. Another limitation is that because this study used real-world data, not all patients underwent biopsy, thereby introducing the possibility that some patients had other cystic lesions, potentially influencing results.

To summarize, in the treatment of SBCs, immediate percutaneous procedures have superior cumulative incidence of healing, but no significant difference in failure or activity restriction time. Watchful waiting often necessitates treatment at a later stage, but remains a possible management strategy if desired. Polidocanol is a superior injectable substance compared with methylprednisolone in terms of cumulative incidence of healing and failure, although it has a more severe complication profile. Lower limb lesions are a negative prognostic factor for healing. After (partial) healing, there

is a significantly reduced risk of failure risk, making this an appropriate timing to discharge from standard follow-up. When choosing a suitable treatment strategy for children with SBC, individual characteristics and considerations should mandate a personalized approach, weighing treatment invasiveness against effectiveness.

Supplementary material

Supplement 1: Modified Neer classification. Supplement 2: Description of excluded patients.

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Data sharing

The datasets generated and analyzed in the current study are not publicly available due to data protection regulations. Access to data is limited to the researchers who have obtained permission for data processing. Further inquiries can be made to the corresponding author.

Ethical review statement

This study was performed in accordance with the 1964 Helsinki Declaration on Ethical Principles for Medical Research Involving Human Participants.

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